Team cyro – Milestone 2: Data Handling & Sketches

Nov 2, 2026 · @Praneeth

Overview

We processed five public datasets into per-question tables and sketched one visualization for each of our eight questions. Two questions changed after we saw what the data can and cannot support (see below).

- **Course:** Information Visualization (CS-GY 6313 B), Fall 2026
- **Team:** cyro (Team 3): Praneeth Yashovardhan Kadem (pk3206), Bhavana Peruri (bp2847), Likhith Vardhan Goruputi (lvg8030)
- **Project:** Congestion Pricing in New York City: What Changed After the Toll?
- **GitHub repository:** github.com/likhith2366/cyro (shared with ctsilva and kimryan0416)

What changed since the proposal

1. **Q2 now uses a second dataset for the "before" period.** The CRZ Vehicle Entries dataset starts on January 5, 2025, the day the toll began, so it has no pre-toll baseline. For the before/after comparison we added *MTA Bridges and Tunnels Hourly Crossings* (2024 vs. 2025, Manhattan-bound crossings). CRZ Entries is still used for patterns since the toll started and for 2025 vs. 2026 changes.
2. **Q3 is reframed as a trend since the toll started.** Excluded-roadway (FDR Drive / West Side Highway) counts also only exist from 2025, so we track their share of all entries over time instead of comparing with before.
3. **Q4 clarifies what "speed" means.** The MTA speeds dataset is computed from taxi and for-hire-vehicle trips, so it measures taxi/FHV travel speeds, not all traffic.
4. **Q6 is split by mode.** Hourly subway ridership counts fare entries by station and cannot tell where riders came from. We compare in-zone station entries 2024 vs. 2025, and use Citi Bike trips (which have both ends) for the "from which neighborhoods" part. The MTA subway origin-destination estimates are a possible extension.
5. **TLC trip records are dropped.** We marked them optional in the proposal; their volume is not needed for any remaining question.

6. **Q8 is a synthesis view** built from the Q4–Q7 outputs rather than a new data source.

Datasets and shared preprocessing

All sources are public. Large tables are aggregated on the server with SoQL (`$select` / `$group`) so we never download millions of raw rows; the scripts in `scripts/` produce every table in this report.

Dataset	ID / access	Period used	Used for
MTA Congestion Relief Zone Vehicle Entries	data.ny.gov `t6yz-b64h`	Jan 2025 – latest	Q1, Q2, Q3
MTA Bridges and Tunnels Hourly Crossings	data.ny.gov `ebfx-2m7v`	2024 – latest	Q2 (pre-toll baseline)
MTA CBD Taxi and FHV Speeds	data.ny.gov `6p29-6xqn`	Oct 2019 – latest	Q4, Q5, Q8
MTA Subway Hourly Ridership: 2020–2024	data.ny.gov `wujg-7c2s`	2024	Q6
MTA Subway Hourly Ridership: Beginning 2025	data.ny.gov `5wq4-mkjj`	2025 – latest	Q6
Citi Bike trip data	monthly ZIPs	Oct 2024, Oct 2025	Q6
NYPD Motor Vehicle Collisions – Crashes	NYC Open Data `h9gi-nx95`	2023 – latest	Q5, Q7, Q8
NTA 2020 boundaries	NYC Open Data `9nt8-h7nd`	static	Q5, Q6, Q8 maps

Steps applied to every dataset

1. **Time:** parse timestamps as America/New_York and add `is_peak` (weekdays 5 a.m.–9 p.m., weekends 9 a.m.–9 p.m.), `is_weekend`, `is_holiday` (US federal holidays) and `year_month`.
2. **Seasonality:** compare the same calendar months across years (e.g., Oct 2024 vs. Oct 2025), never adjacent months.

3. **Space:** label each point `inside` (Congestion Relief Zone), `adjacent` (within 2 km of the zone but outside it) or `elsewhere`. For this milestone the zone is a hand-digitized polygon of Manhattan south of 60th Street (about 100 m accuracy); we will swap in an official boundary file before the final.
4. **Cleaning:** drop rows with missing or zero coordinates; drop Citi Bike trips shorter than 60 seconds or longer than 3 hours; standardize vehicle class and entry-point names.
5. **Gaps:** each script logs date ranges with no data to `data/processed/gaps.csv`, so the final visualizations can mark them.

Domain questions: data, samples and sketches

Each question lists the script that produces it, the transformation, a sample of the processed table (pasted from `output/samples/`) and our sketch.

Q1. When and where do vehicles enter the zone on a typical weekday?

Script: `scripts/q1_q3_crz_entries.py` → `data/processed/q1_weekday_profile.csv`

Transformation

- Keep weekdays (Mon–Fri) that are not federal holidays, Jan 2025 onward.
- Group by entry point (`detection_group`) × hour of day; sum `crz_entries` and `excluded_roadway_entries`.
- Divide by the number of qualifying weekdays to get average entries per weekday hour.

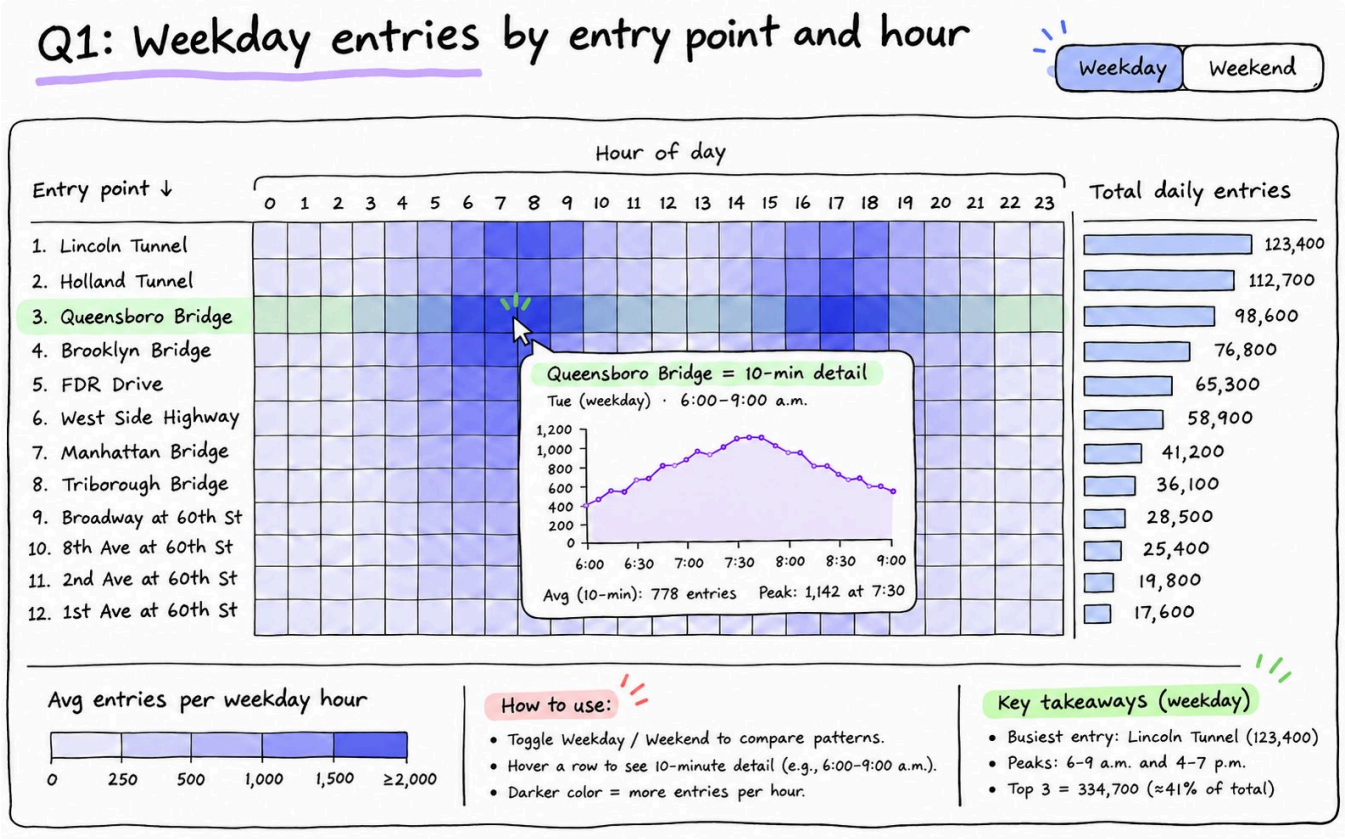
Sample of processed data (first 8 rows of `output/samples/q1.md`)

entry_point	hour	avg_weekday_entries	avg_excluded_entries	n_weekdays
East 60th St	0	2,151.02	0.00	452
East 60th St	1	1,401.57	0.00	452
East 60th St	2	1,008.05	0.00	452
East 60th St	3	968.50	0.00	452
East 60th St	4	1,358.99	0.00	452
East 60th St	5	1,567.46	0.00	452
East 60th St	6	2,416.26	0.00	452

entry_point	hour	avg_weekday_entries	avg_excluded_entries	n_weekdays
East 60th St	7	3,294.71	0.00	452

Full table: 288 rows (12 entry points × 24 hours). Averages over 452 non-holiday weekdays since Jan 5, 2025.

Sketch



Numbers in the sketch are placeholders, not results.

- Heatmap: rows = 12 entry points sorted by daily total; columns = hour 0–23; color = average entries (sequential scale).
- Marginal bar chart on the right: total daily entries per entry point.
- Interaction: weekday/weekend toggle; hovering a row shows its 10-minute profile.

Q2. How have entries changed since the toll started, by vehicle class and time period?

Script: `scripts/q1_q3_crz_entries.py` and `scripts/q2_bt_crossings.py` → `q2_weekly_by_class.csv`, `q2_bt_yoy.csv`

Transformation

- CRZ Entries: group by week × vehicle class × time period (peak / overnight); sum entries. For weeks present in both 2025 and 2026, compute year-over-year % change.
- B&T Crossings (baseline): keep Manhattan-bound crossings at facilities that feed the zone (Hugh L. Carey and Queens Midtown tunnels); sum monthly by vehicle class; compute % change for each 2025 month vs. the same month of 2024.

Sample of processed data (`output/samples/q2_weekly.md`)

week_start	vehicle_class	time_period	entries	entries_same_week_prev_year	pct_change
2026-01-04	1 - Cars, Pickups and Vans	Overnight	343,653	371,003	-7.37
2026-01-04	1 - Cars, Pickups and Vans	Peak	1,475,674	1,506,281	-2.03
2026-01-04	2 - Single-Unit Trucks	Overnight	20,005	20,418	-2.02
2026-01-04	2 - Single-Unit Trucks	Peak	92,511	116,590	-20.65
2026-01-04	3 - Multi-Unit Trucks	Overnight	2,037	3,059	-33.41
2026-01-04	3 - Multi-Unit Trucks	Peak	4,228	7,546	-43.97
2026-01-04	4 - Buses	Overnight	6,316	6,190	2.04
2026-01-04	4 - Buses	Peak	47,397	52,440	-9.62

Full table: 1,116 rows (week × vehicle class × time period). pct_change compares with the same week one year earlier, so it is blank for 2025 weeks; rows shown are the first 2026 week.

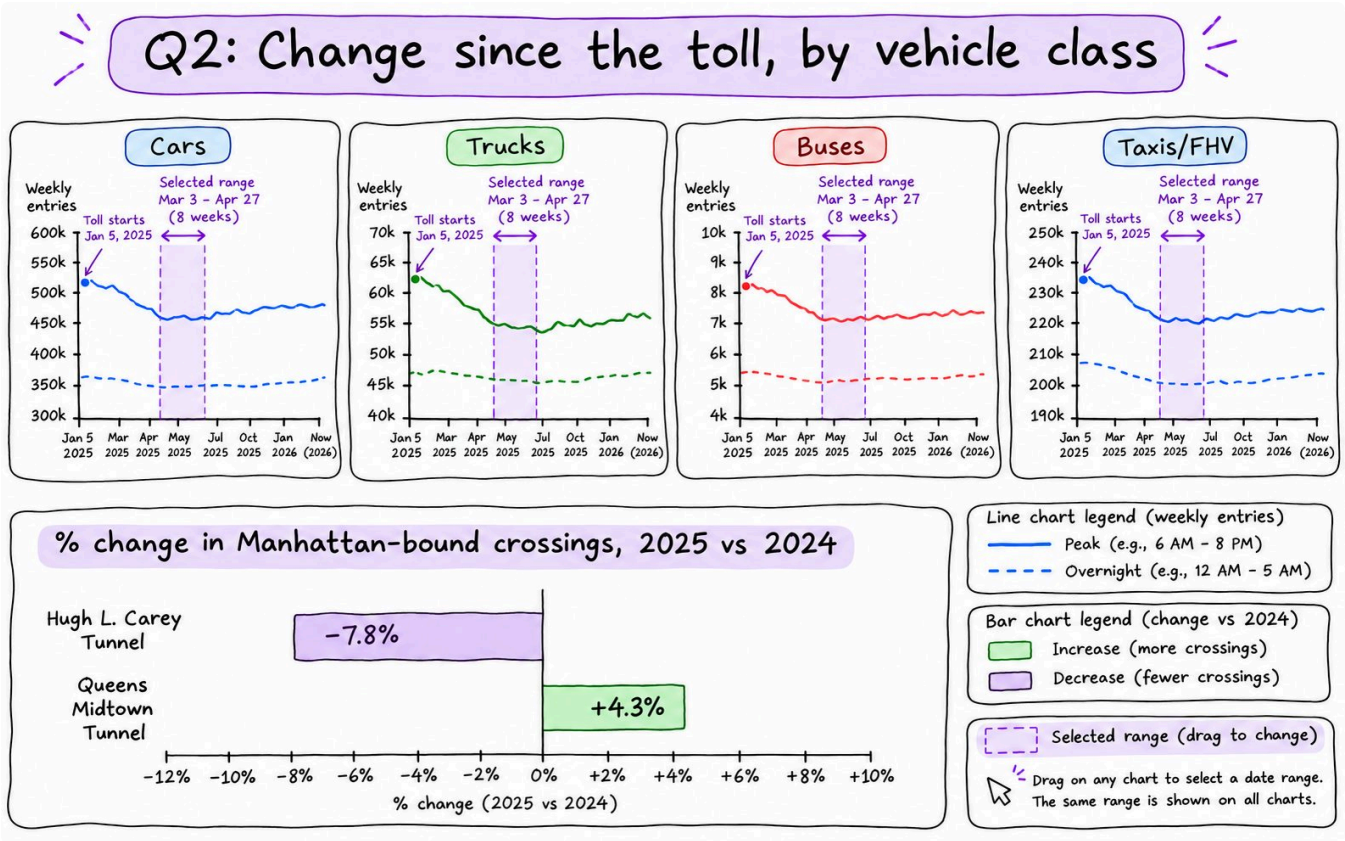
Sample of processed data (`output/samples/q2_bt.md`)

year_month	facility	vehicle_class	crossings	crossings_same_month_2024	pct_change
2025-01	Hugh L. Carey Tunnel	31	803,470	842,662	-4.65
2025-01	Hugh L. Carey Tunnel	32	7,515	8,277	-9.21
2025-01	Hugh L. Carey Tunnel	33	7,651	6,612	15.71
2025-01	Hugh L. Carey Tunnel	34	515	473	8.88

year_month	facility	vehicle_class	crossings	crossings_same_month_2024	pct_change
2025-01	Hugh L. Carey Tunnel	35	61	88	-30.68
2025-01	Hugh L. Carey Tunnel	36	25	18	38.89
2025-01	Hugh L. Carey Tunnel	37	13,816	13,351	3.48
2025-01	Hugh L. Carey Tunnel	38	197	228	-13.60

Full table: 216 rows. Manhattan-bound only (Hugh L. Carey northbound, Queens Midtown westbound). vehicle_class is the MTA B&T class code; for the final we will map codes to the CRZ classes (cars, trucks, buses).

Sketch



Numbers in the sketch are placeholders, not results.

- Small multiples: one panel per vehicle class; x = week; y = weekly entries; two lines per panel (peak, overnight).

- Second view: diverging bars of the 2024→2025 % change per B&T facility, as the before/after evidence.
- Interaction: brush a date range in one panel to zoom all panels.

Q3. Are drivers using the FDR Drive and West Side Highway to avoid the toll?

Script: `scripts/q1_q3_crz_entries.py` → `q3_excluded_share.csv`

Transformation

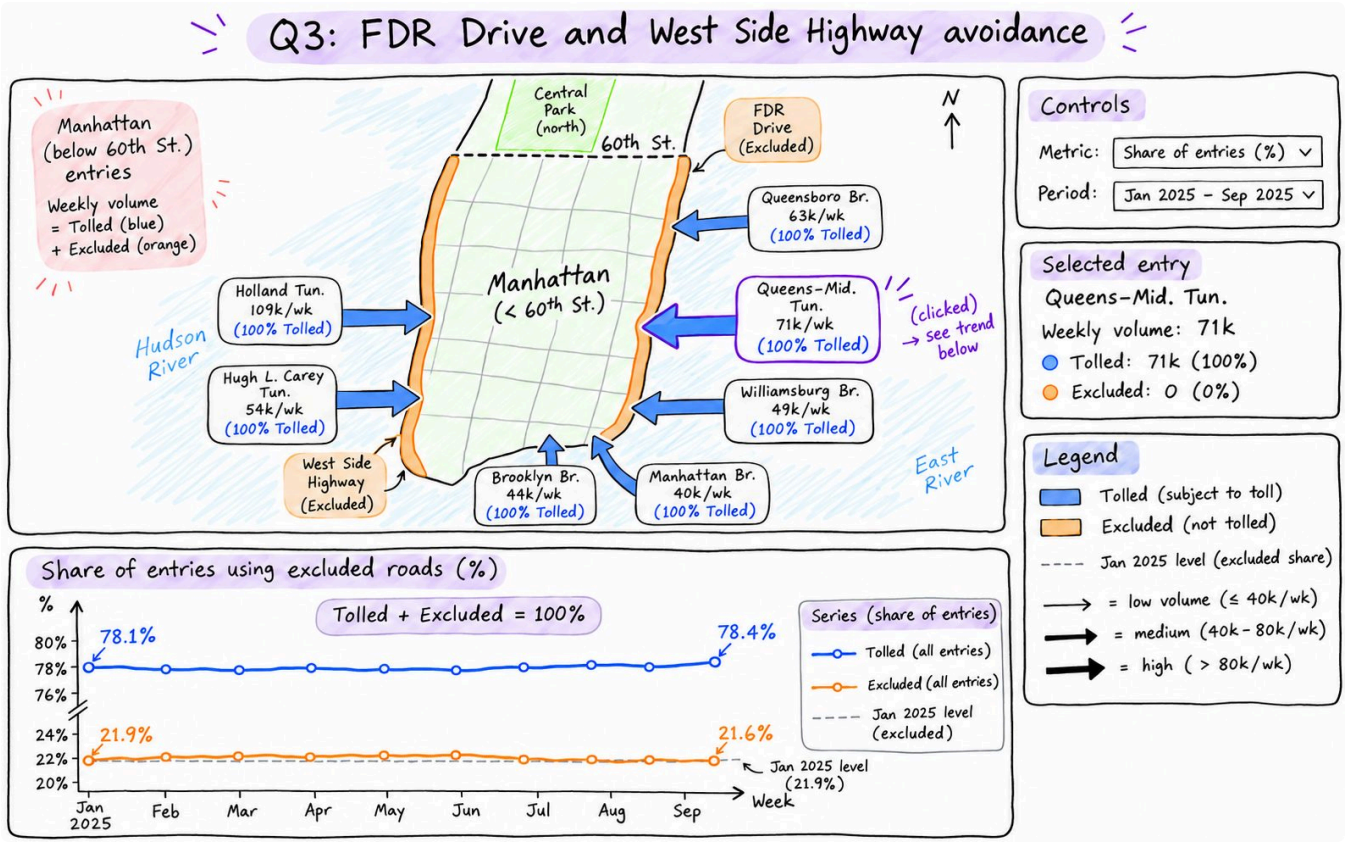
- Group by week × entry point; sum `crz_entries` (tolled) and `excluded_roadway_entries`.
- Compute `excluded_share_pct` = excluded ÷ (tolled + excluded) per week, overall and per entry point.

Sample of processed data (`output/samples/q3.md`)

week_start	entry_point	tolled_entries	excluded_entries	excluded_share_pct
2025-01-05	ALL ENTRY POINTS	3,246,760	424,567	11.56
2025-01-05	Brooklyn Bridge	191,474	139,741	42.19
2025-01-05	FDR Drive at 60th St	353,740	196,740	35.74
2025-01-05	Hugh L. Carey Tunnel	130,933	61,628	32.00
2025-01-05	West Side Highway at 60th St	278,928	26,458	8.66
2025-01-12	ALL ENTRY POINTS	3,346,384	441,770	11.66
2025-01-12	Brooklyn Bridge	199,600	141,900	41.55
2025-01-12	FDR Drive at 60th St	350,955	205,656	36.95

Full table: 1,222 rows; rows with excluded-roadway traffic shown. Finding from the data: excluded entries are not limited to the FDR and West Side Highway. Vehicles entering at the Brooklyn Bridge and Hugh L. Carey Tunnel that stay on the FDR / West Side Highway are also counted as excluded, so the final map will show an excluded share at those entry points too.

Sketch



Numbers in the sketch are placeholders, not results.

- Map of the zone with an arrow at each entry point; arrow width = weekly volume; arrow split into tolled vs. excluded colors.
- Line chart beneath: excluded share over time, with a flat reference line at the first month's share.
- Interaction: click an entry point on the map to filter the line chart.

Q4. Has traffic inside the zone become faster vs. 2019–2024 and vs. the rest of the city?

Script: `scripts/q4_speeds.py` → `q4_speeds_monthly.csv`

Transformation

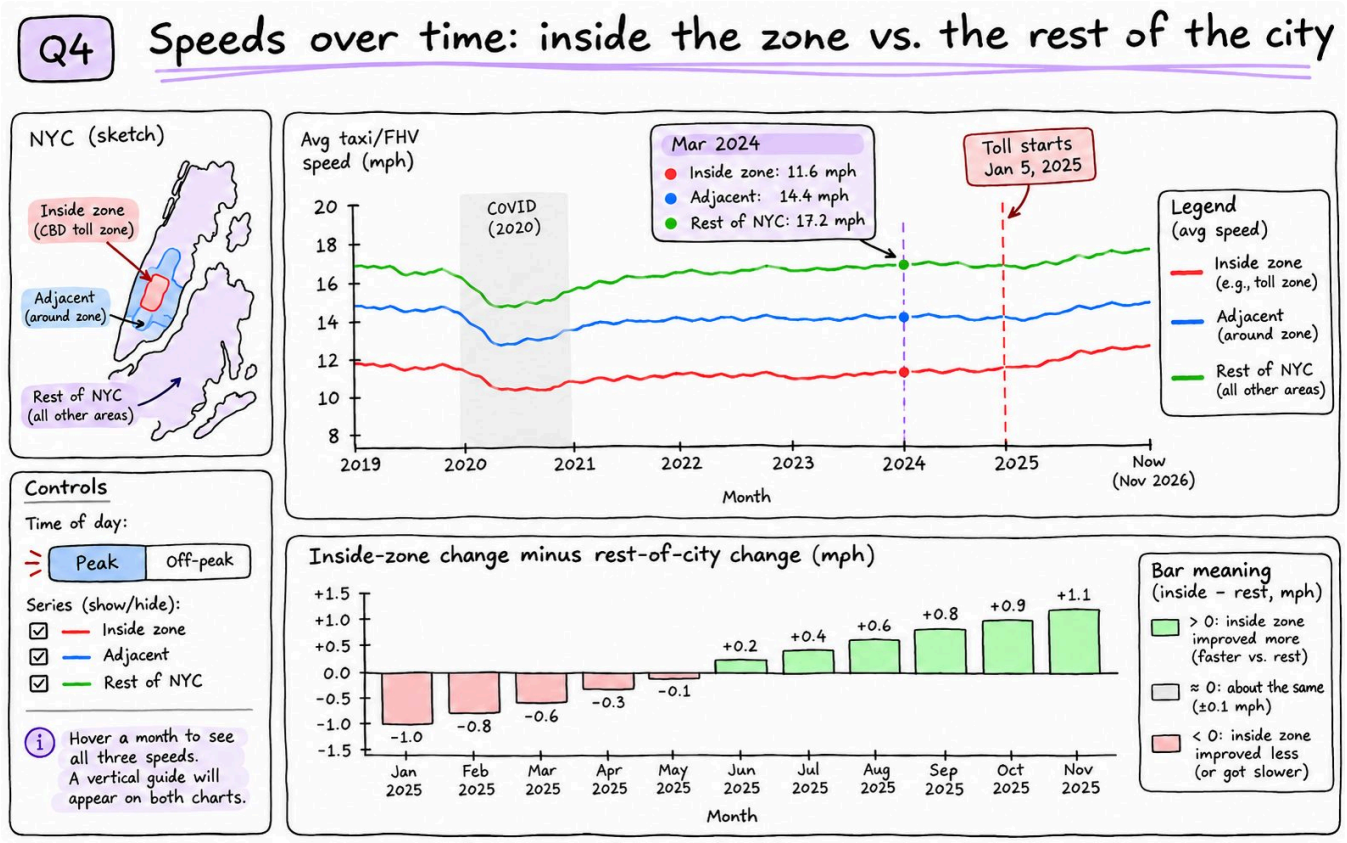
- Monthly average taxi/FHV speed by area (inside zone, adjacent, rest of NYC) and time period, Oct 2019 onward.
- For each 2025+ month, change vs. the same month of 2024, and a simple difference-in-differences: (inside change) − (rest-of-city change).

Sample of processed data (`output/samples/q4.md`)

year_month	area	time_period	avg_speed_mph	speed_same_month_2024	change_mph	did_vs_rest_mph
2025-01	adjacent	all	12.93	12.81	0.12	0.18
2025-01	inside	all	9.66	9.31	0.35	0.41
2025-01	rest_of_nyc	all	17.11	17.17	-0.06	0.00
2025-02	adjacent	all	12.67	12.74	-0.07	-0.00
2025-02	inside	all	9.30	8.99	0.31	0.38
2025-02	rest_of_nyc	all	16.76	16.83	-0.07	0.00
2025-03	adjacent	all	12.81	12.54	0.27	0.12
2025-03	inside	all	9.28	8.76	0.52	0.37

Full table: 213 rows (Oct 2019 onward; 24 rows are 2025+). Areas in the source are "CBD", "CBD Adjacent" and "Greater NYC". Finding from the data: this dataset is monthly with no peak / off-peak split, so the Peak/Off-peak toggle in the sketch needs another source (e.g., MTA CBD bus speeds) or will be dropped.

Sketch



Numbers in the sketch are placeholders, not results.

- Line chart: x = month (2019–now); y = average speed (mph); one line per area; vertical rule at Jan 5, 2025; 2020 lockdown months shaded.
- Panel below: bars of the monthly difference-in-differences (mph), zero baseline.
- Interaction: peak / off-peak toggle; hover shows all three areas for that month.

Q5. Did traffic or crashes get worse just outside the zone?

Script: `scripts/q5_q7_collisions.py` (plus Q4 output) → `q5_nta_crash_change.csv`

Transformation

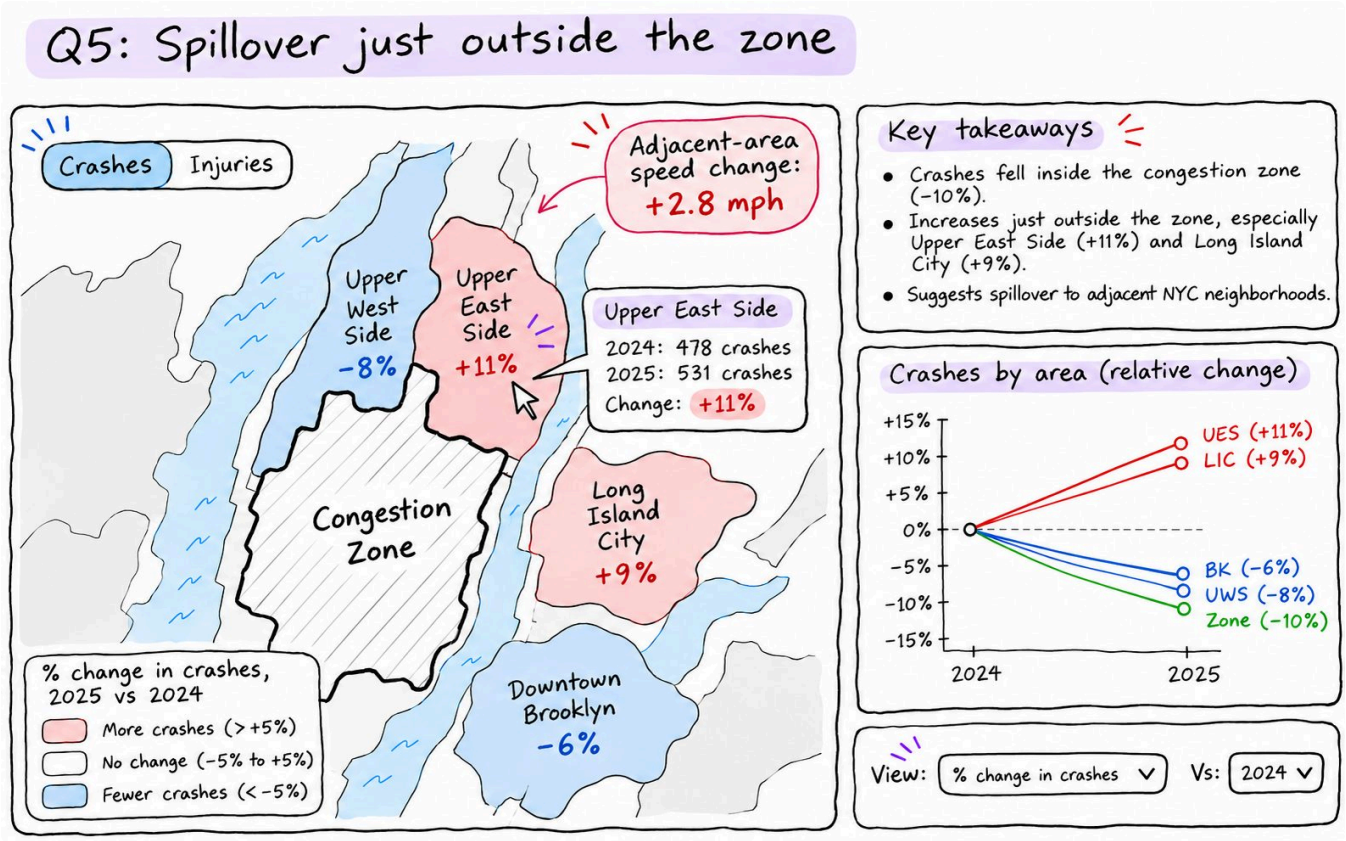
- Collisions 2024 and 2025: drop rows without coordinates; assign each crash to an NTA (spatial join) and to inside / adjacent / elsewhere.
- Per NTA in the adjacent ring: crashes and injuries in 2025 vs. 2024 (same months), % change.
- Adjacent-area speed change comes from the Q4 table.

Sample of processed data (`output/samples/q5.md`)

nta_name	borough	area	crashes_2024	crashes_2025	crash_change_pct	injured_2024	injured_2025
Central Park	Manhattan	adjacent	92	116	26.09	48	73
Brooklyn Heights	Brooklyn	adjacent	227	281	23.79	98	86
Brooklyn Navy Yard	Brooklyn	adjacent	40	46	15.00	22	19
Downtown Brooklyn-DUMBO-Boerum Hill	Brooklyn	adjacent	743	829	11.57	347	332
Long Island City-Hunters Point	Queens	adjacent	335	354	5.67	162	156
Upper East Side-Carnegie Hill	Manhattan	adjacent	381	392	2.89	165	182
Williamsburg	Brooklyn	adjacent	590	591	0.17	271	270
Upper East Side-Lenox Hill-Roosevelt Island	Manhattan	adjacent	488	483	-1.02	261	233

Full table: 30 NTAs inside the zone or within 2 km, sorted by crash change. Same calendar months in both years. 19,960 of 342,342 crashes (5.8%) were dropped for missing coordinates.

Sketch



Numbers in the sketch are placeholders, not results.

- Choropleth of NTAs within 2 km of the zone; color = % change in crashes (diverging, centered on 0); zone outline drawn on top.
- Callout box: adjacent-area speed change (mph) from Q4.
- Interaction: switch metric (crashes / injuries); hover an NTA for its numbers.

Q6. Did more people take the subway or Citi Bike into the zone, and from where?

Scripts: `scripts/q6_subway.py` → `q6_subway_monthly.csv`; `scripts/q6_citibike.py` → `q6_citibike_origins.csv`

Transformation

- Subway: monthly fare entries per station complex (server-side sum); label stations inside / adjacent / elsewhere; compare each 2025 month with the same month of 2024.
- Citi Bike: October 2024 and October 2025 trips; keep trips that end inside the zone; assign the start station to an NTA; count trips per origin NTA and compute % change.

Sample of processed data (`output/samples/q6_subway.md`)

year_month	station_complex	area	nta_name	entries	entries_same_month_2024	pct_change
2025-01	1 Av (L)	inside	East Village	583,974	472,542	23.58
2025-01	14 St-Union Sq (4,5,6,L,N,Q,R,W)	inside	Midtown South-Flatiron-Union Square	1,840,647	1,709,309	7.68
2025-01	14 St/6 Av (1,2,3,L,F,M)	inside	Chelsea-Hudson Yards	877,174	793,798	10.50
2025-01	14 St/8 Av (A,C,E,L)	inside	Chelsea-Hudson Yards	897,626	896,864	0.08
2025-01	18 St (1)	inside	Chelsea-Hudson Yards	151,124	137,379	10.01
2025-01	2 Av (F)	inside	East Village	349,388	329,873	5.92
2025-01	23 St (1)	inside	Chelsea-Hudson Yards	272,058	239,663	13.52
2025-01	23 St (C,E)	inside	Chelsea-Hudson Yards	411,952	398,496	3.38

Full table: 14,108 station-months (2024 from wujg-7c2s, 2025+ from 5wq4-mkjj); 1,386 rows are in-zone stations with a 2024 comparison. station_complex_id and borough columns omitted here for width.

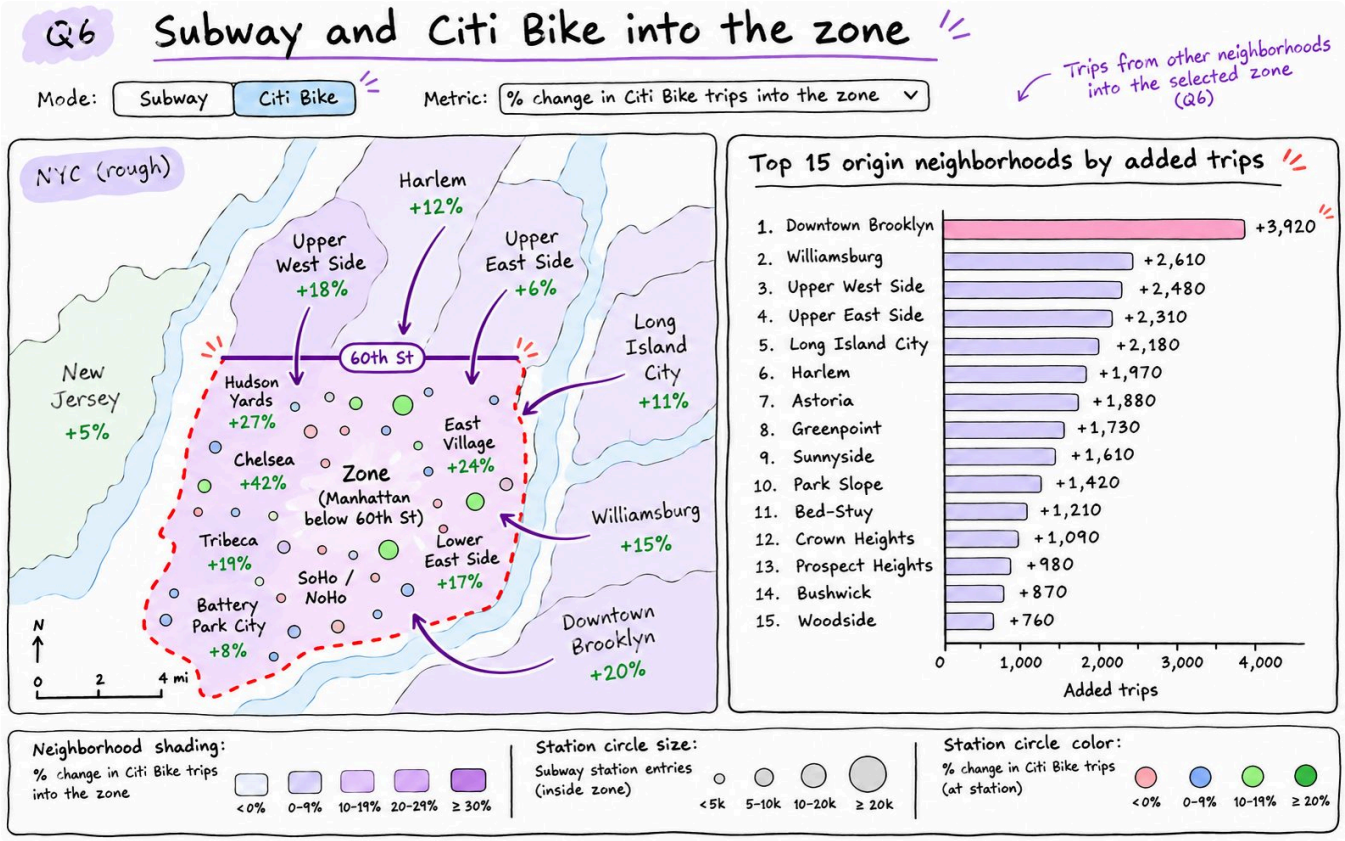
Sample of processed data (`output/samples/q6_citibike.md`)

origin_nta	origin_borough	origin_area	trips_into_zone_202410	trips_into_zone_202510	pct_change	added_trips
Murray Hill-Kips Bay	Manhattan	inside	127,771	130,967	2.50	3,196
Manhattanville-West Harlem	Manhattan	elsewhere	1,141	1,275	11.74	134
Astoria Park	Queens	elsewhere	79	155	96.20	76
Sunnyside Yards (South)	Queens	elsewhere	107	150	40.19	43
Kensington	Brooklyn	elsewhere	130	161	23.85	31
East Flatbush-Rugby	Brooklyn	elsewhere	39	57	46.15	18
Washington Heights (North)	Manhattan	elsewhere	543	560	3.13	17

origin_nta	origin_borough	origin_area	trips_into_zone_202410	trips_into_zone_202510	pct_change	added_trips
Randall's Island	Manhattan	elsewhere	307	320	4.23	13

Full table: 122 origin NTAs, sorted by added trips. 2.46 M (Oct 2024) and 2.28 M (Oct 2025) valid trips ended in the zone, so total trips into the zone fell 7%; only a few origins grew.

Sketch



Numbers in the sketch are placeholders, not results.

- Map: NTAs colored by % change in Citi Bike trips into the zone; subway stations inside the zone as circles sized by entries and colored by % change.
- Side bar chart: top 15 origin NTAs by added trips.
- Interaction: mode toggle (subway / Citi Bike); clicking an NTA draws its flow line into the zone.

Q7. Have crashes and injuries inside the zone gone down compared with the rest of the city?

Script: `scripts/q5_q7_collisions.py` → `q7_monthly_by_area.csv`

Transformation

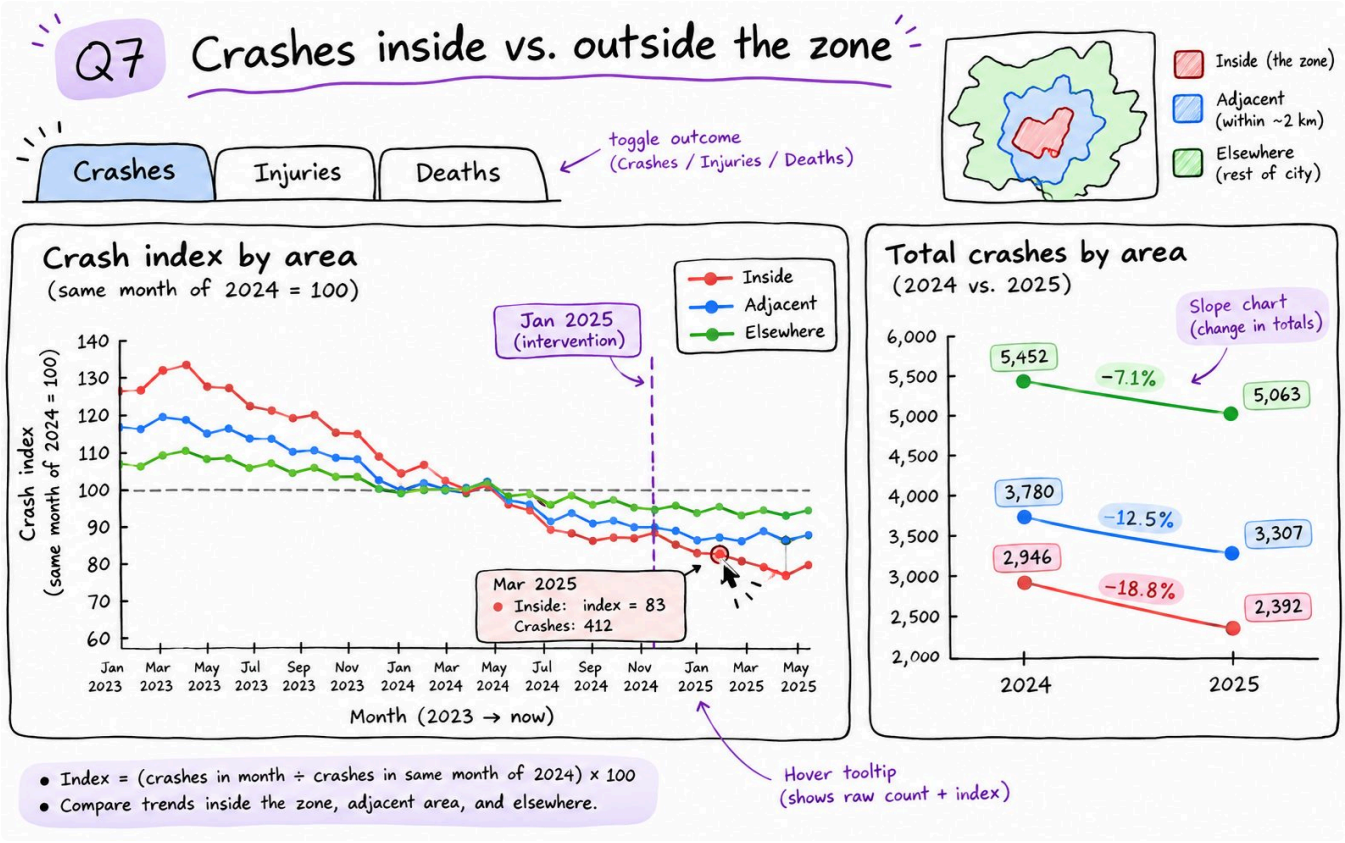
- Monthly crashes, injuries and deaths per area (inside / adjacent / elsewhere), 2023 onward.
- Index each month to the same month of 2024 = 100 so the three areas share one scale.

Sample of processed data (`output/samples/q7.md`)

year_month	area	crashes	persons_injured	persons_killed	crash_index_2024_100
2025-01	adjacent	334	142	1	94.35
2025-01	elsewhere	5,474	3,113	14	88.78
2025-01	inside	458	204	1	83.58
2025-02	adjacent	277	126	3	97.88
2025-02	elsewhere	5,067	2,651	12	90.05
2025-02	inside	440	224	2	87.13
2025-03	adjacent	359	166	1	120.07
2025-03	elsewhere	5,839	3,434	14	90.57

Full table: 135 rows (2023 onward); 63 rows are 2025+. Index = crashes ÷ crashes in the same month of 2024 × 100.

Sketch



Numbers in the sketch are placeholders, not results.

- Indexed line chart: x = month; y = crash index (2024 = 100); one line per area; reference line at 100.
- Slope chart beside it: 2024 → 2025 totals per area for crashes, injuries and deaths.
- Interaction: metric selector; hover shows raw counts.

Q8. Which areas gained the most, and which carried the costs?

Script: `scripts/q8_summary.py` (joins Q4–Q7 outputs) → `q8_nta_summary.csv`

Transformation

- One row per NTA: crash change %, Citi Bike into-zone trip change %, subway entry change % (stations in the NTA), and the area-level speed change.
- Each metric is signed so that positive = benefit (e.g., fewer crashes = positive).

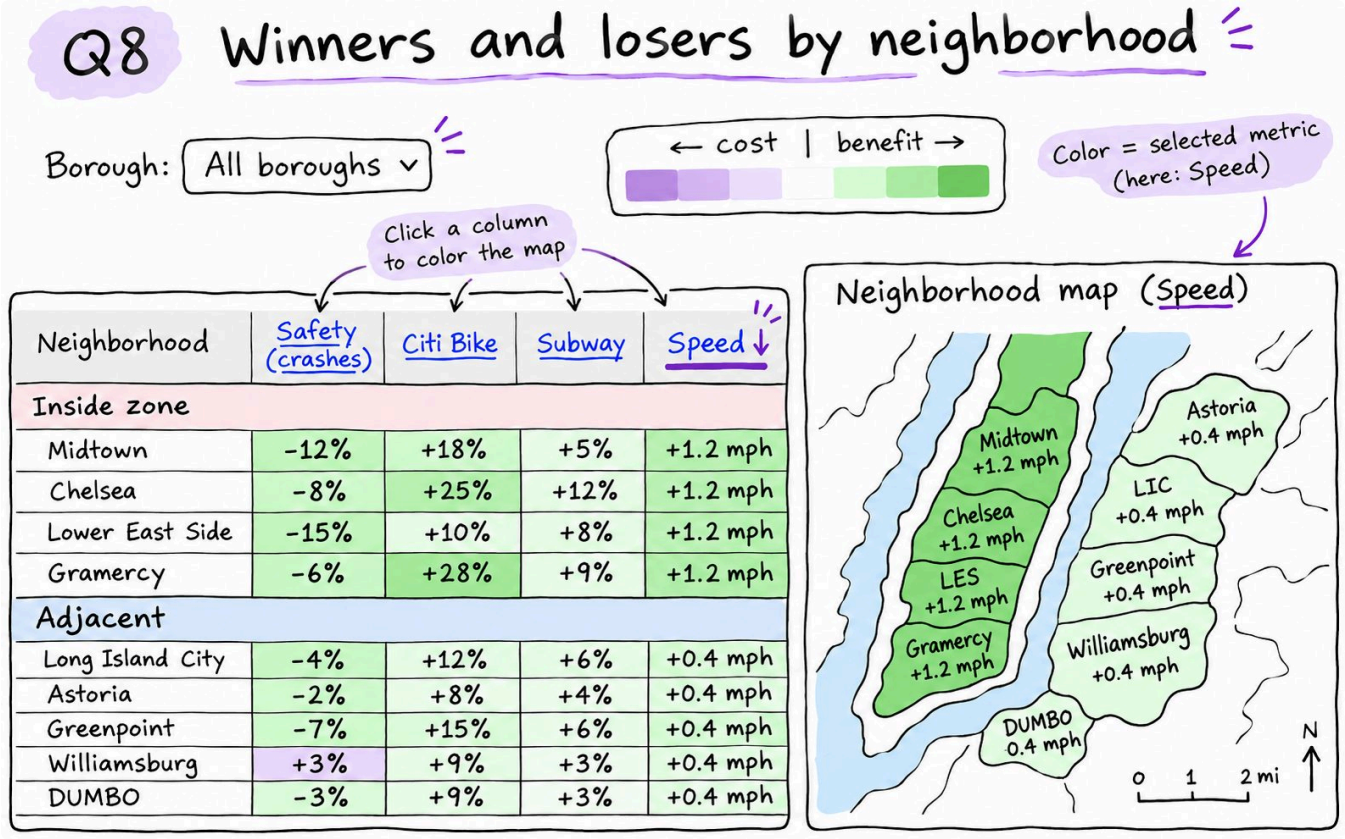
Sample of processed data (`output/samples/q8.md`)

nta_name	borough	area	crash_change_pct	citibike_change_pct	subway_change_pct	speed_change_mph
Brooklyn Heights	Brooklyn	adjacent	23.79	-10.29	-3.27	-0.14
Brooklyn Navy Yard	Brooklyn	adjacent	15.00	-0.63	(no station)	-0.14

nta_name	borough	area	crash_change_pct	citibike_change_pct	subway_change_pct	speed_change_mph
Central Park	Manhattan	adjacent	26.09	-26.21	0.24	-0.14
Downtown Brooklyn-DUMBO-Boerum Hill	Brooklyn	adjacent	11.57	-12.08	4.25	-0.14
Greenpoint	Brooklyn	adjacent	-12.42	-6.70	21.58	-0.14
Long Island City-Hunters Point	Queens	adjacent	5.67	-3.49	8.85	-0.14
Queensbridge-Ravenswood-Dutch Kills	Queens	adjacent	-21.31	-11.57	15.43	-0.14
South Williamsburg	Brooklyn	adjacent	-13.01	-7.75	6.85	-0.14

Full table: 30 NTAs inside the zone or within 2 km. speed_change_mph is the area-level 2025 average from Q4 (same value for every NTA in an area). Benefit-signed columns (safety_benefit = −crash_change_pct, etc.) are in the CSV.

Sketch



Numbers in the sketch are placeholders, not results.

- Matrix heatmap: rows = NTAs grouped by area; columns = metrics; color = signed change (diverging: benefit vs. cost).
- Linked map: clicking a column colors the NTA map by that metric.
- Interaction: sort rows by any column; filter by borough.

Data gaps and limitations

- **No pre-toll baseline in CRZ Entries or excluded-roadway counts.** Before/after claims rely on B&T crossings (Q2) and speeds (Q4).
- **Correlation, not causation.** Return-to-office trends, weather, construction and fare changes all coincide with 2025. We compare against adjacent and citywide areas and label differences as associations.
- **Speeds are taxi/FHV-based** and may not reflect trucks or private cars.
- **Subway entries count where riders start, not where they go.** In-zone station entries are a proxy for zone travel.
- **The zone polygon is approximate** (hand-digitized, about 100 m) until we load an official boundary.
- **Collision coordinates are missing for some crashes**; we report how many rows were dropped per month.

AI use statement

Per the course AI policy:

- We used Claude (Anthropic) to draft the structure and text of this report, the README, and the data-processing scripts in `scripts/`. We reviewed, ran and edited all of it; every sample table above is the output of those scripts run by us.
- **All sketches are our own work.** No AI tool was used to create or edit them.
- Earlier, Claude helped brainstorm the topic and draft parts of the proposal (as noted in Milestone 1).